Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 08)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 1 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 1 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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0 2 Figure 1 shows a simplified energy budget for the Earth and its atmosphere. Values are percentages of incoming solar radiation.

Figure 1

Energy flow	Percentage of incoming solar radiation / %
Reflected by clouds, dust and atmosphere	25
Reflected by the Earth's surface	5
Absorbed by the atmosphere	20
Absorbed by the Earth's surface	50
Infrared emitted by the surface and absorbed by greenhouse gases	95
Infrared re-emitted by the atmosphere back to the surface	85
Infrared emitted to space	70

0 2 . 1 Use Figure 1 to calculate the planetary albedo of the Earth.

[1 mark]

Answer ____ %

0 2 . 2 Describe how each of the following wavelengths of electromagnetic radiation behaves in the atmosphere.

[6 marks]

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Ultraviolet (UV) .....

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Visible light .....

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Infrared (IR) .....

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0 2 . 3 Explain why the amount of infrared re-emitted to the surface (85%) is larger than the amount emitted to space (70%).

[2 marks]

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0 2 . 4 Explain the term **thermal stratification** of the atmosphere.

[1 mark]

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0 3 In 2020 the tanker MV Wakashio ran aground on a coral reef off Mauritius and leaked about 1000 tonnes of fuel oil into a shallow lagoon near mangroves and a marine park.

0 3 . 1 Explain why inflatable booms were deployed immediately, and give **one** limitation of booms.

[3 marks]

0 3 . 2 Describe how skimmers and absorbent materials remove oil, and state how the collected oil is disposed of.

[3 marks]

0 3 . 3 Explain why dispersants were **not** used in the lagoon.

[2 marks]

0 3 . 4 Explain how bioremediation could help remove the oil that remained on the mangrove sediments, and suggest how it could be accelerated.

[2 marks]

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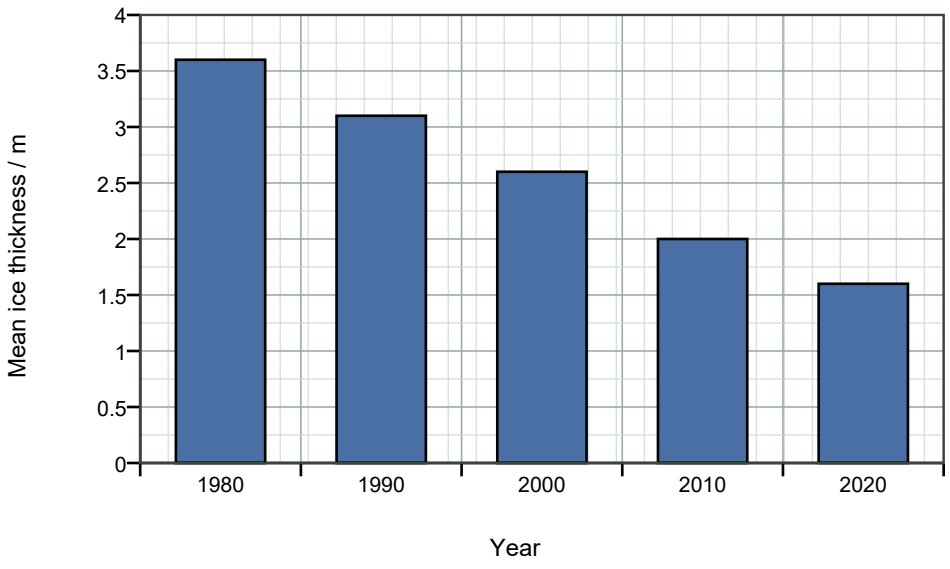
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0 4 Table 1 shows the area of Arctic sea ice in September (the annual minimum) in selected years. Figure 1 shows the mean thickness of Arctic sea ice in the same years.

Table 1

Year	1980	1990	2000	2010	2020
September sea ice area / million km^2	7.7	6.2	6.3	4.9	3.9

Figure 1



0 4 . 1 Calculate the percentage decrease in September sea ice area between 1980 and 2020.

[2 marks]

Answer ..... %

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0 4 . 2 Explain why the loss of sea ice does **not** directly raise sea level, but the loss of land ice does.

[2 marks]

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0 4 . 3 Explain how the changes shown in Table 1 and Figure 1 could lead to further warming.

[3 marks]

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0 4 . 4 Explain **two** ways in which rising temperatures cause sea level to rise.

[2 marks]

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0 4 . 5 Explain why the break-up of an ice shelf can increase the rate at which land ice flows into the sea.

[1 mark]

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0 5 . 1 Describe the main features of the Montreal Protocol (1987).

[3 marks]

0 5 . 2 For each of the following CFC uses, give the alternative process or material now used and one disadvantage of the alternative.

[4 marks]

Aerosol propellant

Refrigerant

0 5 . 3 Describe how waste CFCs from old refrigerators are disposed of safely.

[2 marks]

0 5 . 4 State **one** reason why the ozone layer is not expected to recover fully for many decades.

[1 mark]

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0 6 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

0 6 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

0 6 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

0 6 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

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0 7 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 7 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 7 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 8 Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

0 8 . 1 Suggest how the inorganic mercury was converted to methyl mercury.

[2 marks]

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0 8 . 2 Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.

[3 marks]

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0 8 . 3 Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.

[2 marks]

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0 8 . 4 Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.

[3 marks]

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0 9 . 1 Explain why phosphorus is more likely than nitrogen to be a limiting factor on biological productivity.

[3 marks]

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0 9 . 2 Explain how mycorrhizal fungi increase phosphate uptake by plants.

[2 marks]

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0 9 . 3 Describe how human activities have altered the phosphorus cycle.

[3 marks]

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0 9 . 4 Explain **two** strategies for the sustainable management of phosphorus supplies.

[2 marks]

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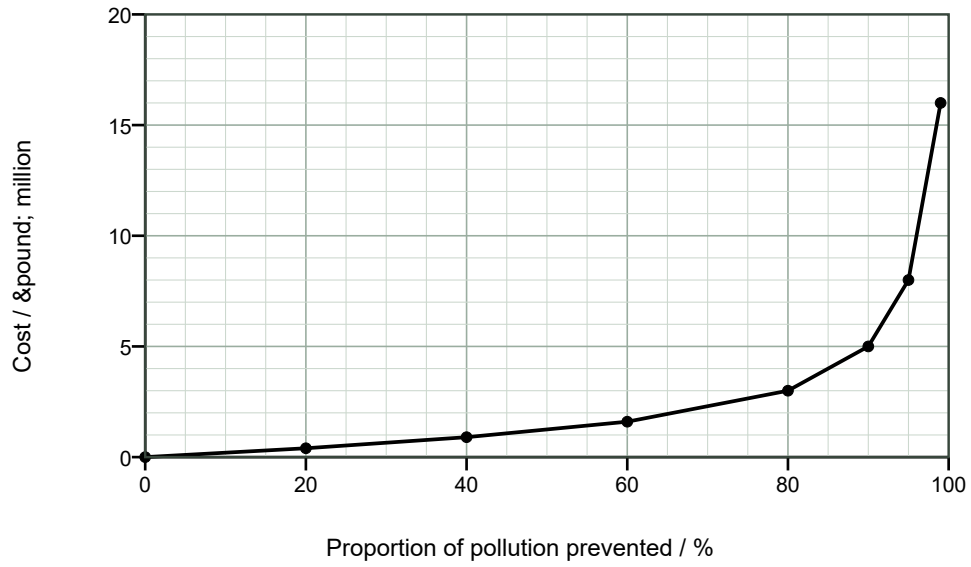
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1 0 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer [ ] £ million

1 0 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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1 0 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

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1 0 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1 1 . 2 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS